Spearman's rank correlation coefficient

1

a Pearson's correlation coefficient because there is a clear upward trend.

b Spearman's rank coefficient because the trend is not linear.

2

a

i

x	3	4	5	6	7	8	9
Rank of x	1	2	3	4	5	6	7
y	7	7.4	7.88	7.64	7.72	8.2	7.32
Rank of y	1	3	6	4	5	7	2

ii $r_s = 0.36$

iii Weak, positive correlation

b

i

x	2	6	7	14	17	22
Rank of x	1	2	3	4	5	6
y	2.0	2.4	2.0	0.6	0.9	0.2
Rank of y	4.5	6	4.5	2	3	1

ii $r_s = -0.84$

iii Strong, negative correlation

c

i

x	1	4	7	10	13	16	19
Rank of x	1	2	3	4	5	6	7
y	-1	-1.7	-1.7	-1.84	-1.91	-1.98	-2.05
Rank of y	7	5.5	5.5	4	3	2	1

ii $r_s = -0.99$

iii Strong, negative correlation

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d

i

x	4	6	8	13	17	21
Rank of x	1	2	3	4	5	6
y	0.4	0.9	0.6	1.7	2.4	1.7
Rank of y	1	3	2	4.5	6	4.5

ii $r_s = 0.84$

iii Strong, positive correlation

3

a $r_p = 0.958$

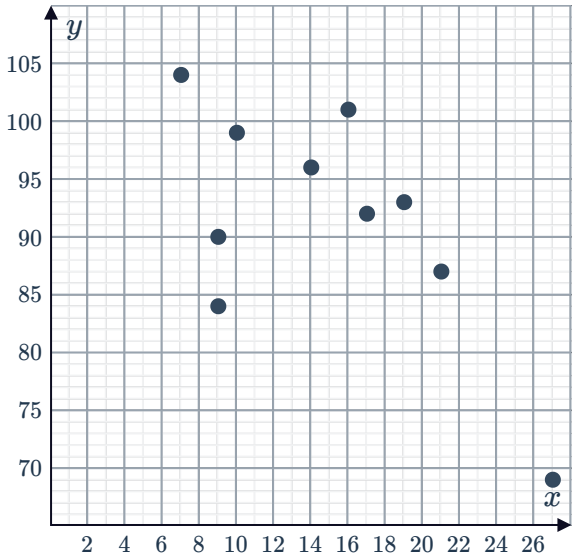
b $r_s = 1.00$

c Very strong positive correlation.

Applications

4

a



b

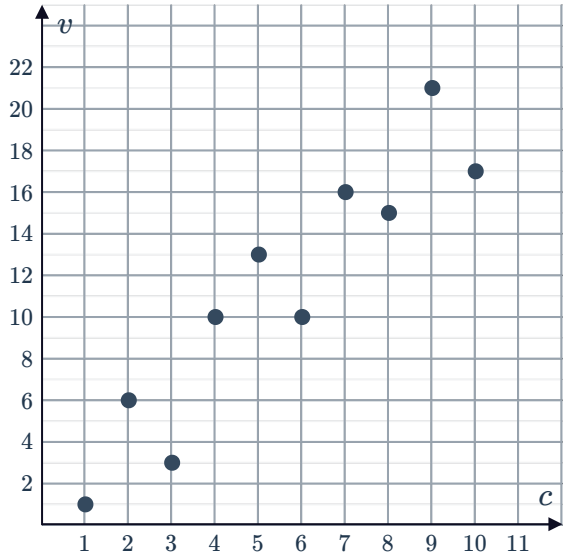
Age when first spoke, x	14	27	9	9	16	21	17	10	7	19
Rank of x	5	10	2.5	2.5	6	9	7	4	1	8
Aptitude test results, y	96	69	84	90	101	87	92	99	104	93
Rank of y	7	1	2	4	9	3	5	8	10	6

c $r_s = -0.444$

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5

a



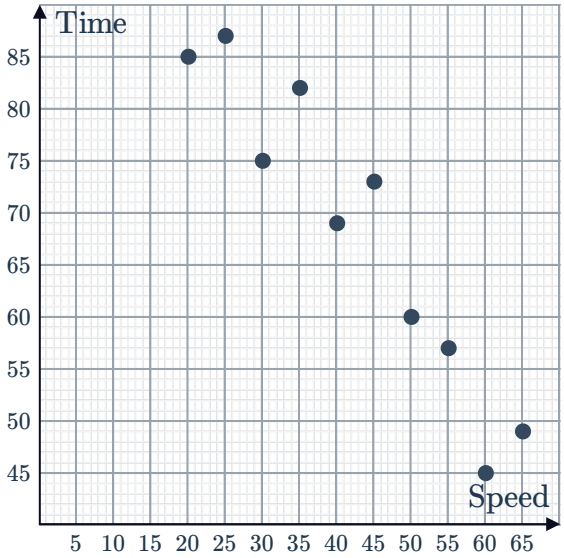
b

Current, c	1	2	3	4	5	6	7	8	9	10
Rank of c	1	2	3	4	5	6	7	8	9	10
Voltage, v	1	6	3	10	13	10	16	15	21	17
Rank of v	1	3	2	4.5	6	4.5	8	7	10	9

c $r_s = 0.942$

6

a



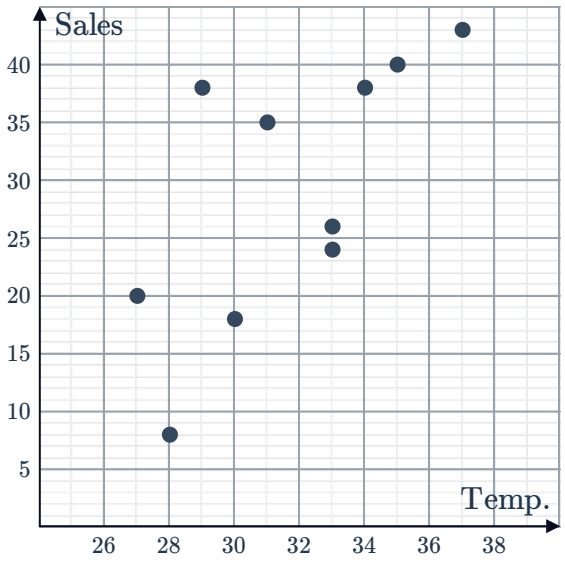
b $r_s = -0.952$

c Strong, negative, linear correlation

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7

a

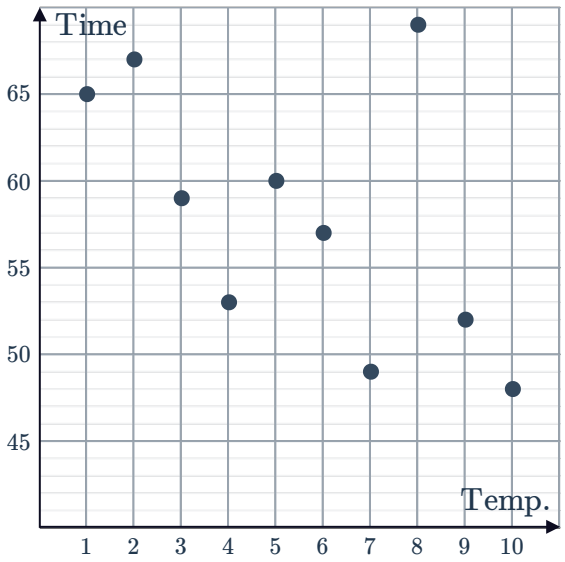


b $r_s = 0.76$

c The sales tend to increase.

8

a



b 10

c $r_s = -0.564$

d Moderate, negative, linear correlation

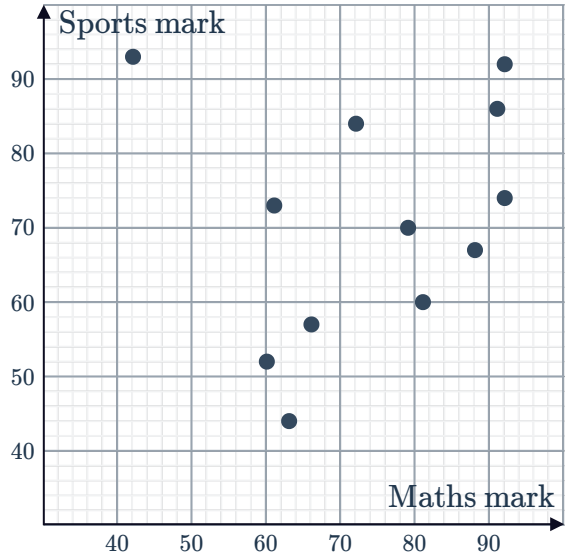
e 8°C, 69 seconds

f $r_s = -0.883$

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9

a



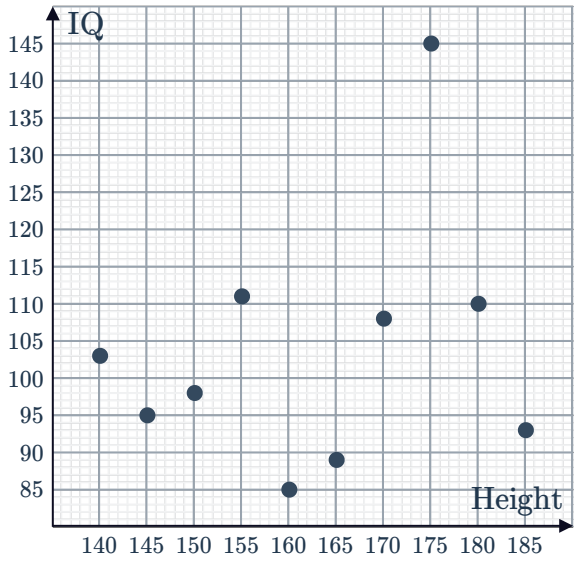
b Weak, positive, linear correlation

c Student 10's

d $r_s = 0.670$

10

a



b Not correlated

c Height $= 175$ cm

d $r_s = -0.05$

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